Girls Build Bots — Cadette App Development

Badge-aligned workshop guide for robotics-team-led outreach

Level: Cadettes (Grades 6–8) · **Badge:** Coding for Good — App Development **Series:** Coding for Good (Coding Basics · Digital Game Design · App Development) **Format:** One 2-hour session (2.5 with the App Inventor option) · **Ratio:** 1 mentor per 5–6 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — the core session runs on paper; the optional build uses free MIT App Inventor.

> **Troop Leaders:** This curriculum is designed to *align with* the Cadette Coding for Good badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: coding *for good*

This is the capstone "for good" badge: Cadettes design an app for **social good** — a real community problem, a real user, a defensible solution. The standard is professional: user research before design, design before code.

Vocabulary: user research, problem statement, MVP (minimum viable product), wireframe, user flow, prototype, accessibility.

App Development (2 hours; +30 min with App Inventor build)

Badge theme: Planning and prototyping an app for social good — from user needs through interface design.

Materials

- Paper wireframe templates (phone outlines, 5–6 per team) + sticky notes, markers
- "Community problem" prompt cards (food-pantry donations coordination; new-student buddy matching; neighborhood tool-lending; elder tech-help scheduling; lost-pet alerts) — or teams bring their own real problem
- User-interview scripts; mentors prepped to role-play users
- Optional: laptops with **MIT App Inventor** (appinventor.mit.edu) for a working one-screen build

Activity 1 — Apps for Good, Deconstructed (15 min)

Examine two apps built for social good (a food-rescue app, a crisis-help finder — describe or screenshot). For each: who's the user, what problem, and **what's the smallest thing it**

does that helps? Introduce the **MVP**: real developers ship the minimum that solves the problem, then grow it. Big hearts, small first versions.

Activity 2 — User Research (25 min)

Teams of 2–3 pick a community problem (a card, or better, a real one they know). A mentor role-plays the user; teams run a real interview: walk me through the problem, what have you tried, what would "solved" look like? Deliverable: a **problem statement** — *[User] needs a way to [goal] because [barrier]* — plus the top two must-haves from the user's own words. Rule: no design talk until research is done.

Activity 3 — User Flow + Wireframes (40 min)

Design the MVP on paper: 1. **User flow first:** sticky-note map of the user's path — open app → do the core thing → confirmation. Three to five steps, no more. 2. **Wireframe each step:** one paper screen per sticky note. Core action gets the biggest button. 3. **Accessibility pass** — new at this level: big-enough touch targets, readable text, does it work for a user who is older, in a hurry, or new to phones? Design for the *whole* community. 4. Feature-cut round: anything not serving the problem statement gets removed. Cutting is designing.

Activity 4 — Click Test + Pitch (25 min) + Wrap (15 min)

The mentor-user runs a **click test** on the paper prototype: they think aloud while "tapping"; the team silently swaps screens and takes notes, then makes one fix. Then each team gives a 90-second pitch: the problem, the user, the MVP, and what they'd build next with more time.

Optional +30 min: in MIT App Inventor, build the welcome screen + core button as a real running app on a phone or emulator — one working screen beats five imagined ones.

> **Robot moment (optional):** Share for 60 seconds how the team scopes its robot each season to an MVP — the minimum machine that scores — before adding stretch features. Same discipline, bigger build.

Completion Checklist — App Development

By the end of the session, to earn the badge, each Scout can: - [ ] Explain what an MVP is and why developers start there - [ ] Conduct user research and write a problem statement from it - [ ] Design a user flow and wireframes where every screen serves the user - [ ] Run a click test, iterate, and pitch the app's purpose and next steps

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Mentor Guide (for the robotics team)

- **Treat Cadettes as a junior product team.** Real roles emerge — researcher, designer, presenter. Let them.
- **You don't need a robot for this.** The "robot moment" is a 60-second add-on; the optional App Inventor build is the better stretch if you have time.
- **Play a demanding user.** Have opinions, get confused, contradict yourself slightly — real users do.

- **Enforce research-before-design.** Teams that sketch during interviews get gently sent back to listening.
- **The accessibility pass is non-negotiable.** It's the "for good" made concrete: an app that excludes users isn't solving the community's problem.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Cadette Coding for Good badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.